

Mathematical Foundations of Reinforcement Learning and Online Decision Making

Lecture 5

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How to estimate the value function when we don't know the model?

Criterion 1: Discounted Problem

$$V(s) = \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t R(S_t) \mid S_0 = s \right]$$

Criterion 2: Stochastic Shortest Path Problem

$$V(s) = \mathbb{E} \left[\sum_{t=0}^{\infty} R(X_t) \mid S_0 = s \right]$$

Criterion 3: Average Cost per Stage Problem

$$V(s) = \mathbb{E} \left[\limsup_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T R(S_t) \mid S_1 = s \right]$$

Overview

- 1 Monte Carlo Estimation
- 2 Monte Carlo Estimation for Value Estimation
- 3 Importance Sampling
- 4 TD-Learning in MRP

Monte Carlo Estimation

Problem

We have a random variable v with distribution ρ and unknown mean $\mathbf{E}[v] = m$. If we only have N samples

$$v_1, \dots, v_N \sim \rho$$

how can we estimate m ?

Monte Carlo Estimation

Problem

We have a random variable v with distribution ρ and unknown mean $\mathbf{E}[v] = m$. If we only have N samples

$$v_1, \dots, v_N \sim \rho$$

how can we estimate m ?

We consider the estimator

$$M_N = \frac{1}{N} \sum_{k=1}^N v_k$$

If we get a new sample v_{N+1} then we can update M_N iterative as:

$$M_{N+1} = M_N + \frac{1}{N+1}(v_{N+1} - M_N)$$

Monte Carlo Estimation - IID Case

IID Samples

Suppose that $v_1, \dots, v_N \sim \rho$ are independent with mean m and variance σ^2 . Consider the estimator

$$M_N = \frac{1}{N} \sum_{k=1}^N v_k$$

Monte Carlo Estimation - IID Case

IID Samples

Suppose that $v_1, \dots, v_N \sim \rho$ are independent with mean m and variance σ^2 . Consider the estimator

$$M_N = \frac{1}{N} \sum_{k=1}^N v_k$$

Unbiased:
$$\mathbf{E}[M_N] = \frac{1}{N} \sum_{k=1}^N \mathbb{E}v_k = m$$

Monte Carlo Estimation - IID Case

IID Samples

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$$M_N = \frac{1}{N} \sum_{k=1}^N v_k$$

Unbiased:
$$\mathbf{E}[M_N] = \frac{1}{N} \sum_{k=1}^N \mathbb{E}v_k = m$$

MSE:
$$\mathbf{Var}[M_N] = \frac{1}{N^2} \sum_{k=1}^N \sigma^2 = \frac{\sigma^2}{N}$$

Monte Carlo Estimation - IID Case

Law of large numbers

By Chebychev's inequality we get that for all $\epsilon > 0$

$$\mathbf{P}(|M_N - m| \geq \epsilon) \leq \frac{\sigma^2}{N\epsilon^2}$$

We can conclude that the estimator M_n is consistent, i.e., it converges in probability to m .

Monte Carlo Estimation - non-IID Case

Dependent Samples

Suppose that $v_1, \dots, v_N \sim \rho$ are ~~independent~~ with mean m and variance σ^2 . Consider the estimator

$$M_N = \frac{1}{N} \sum_{k=1}^N v_k$$

Monte Carlo Estimation - non-IID Case

Dependent Samples

Suppose that $v_1, \dots, v_N \sim \rho$ are ~~independent~~ with mean m and variance σ^2 . Consider the estimator

$$M_N = \frac{1}{N} \sum_{k=1}^N v_k$$

Unbiased: $\mathbf{E}[M_N] = \frac{1}{N} \sum_{k=1}^N \mathbb{E} v_k = m$

MSE: $\mathbf{Var}[M_N] = \frac{\sigma^2}{N} + \sum_{i,j=1, i \neq j}^N \mathbf{Cov}(v_i, v_j)$

Monte Carlo Estimation - non-IID Case

General weighted estimators

Suppose that $v_1, \dots, v_N \sim \rho$ are possibly dependent with mean m and variance σ^2 . Consider the estimator

$$M_N = \sum_{k=1}^N w_k v_k$$

where $w_k \in [0, 1]$ and $w_1 + w_2 + \dots + w_N = 1$.

- Are there better estimators than using the average, i.e., setting $w_k = 1/N$?
- Do all samples contain some new information? or should we disregard some samples, i.e., put $w_k = 0$ for some k .

Monte Carlo Estimation - non-IID Case

Example 1 from Bersekas & Tsitkil p.182

Suppose that we have 3 samples $v_1, v_2, v_3 \sim \rho$ with mean m and variance 1. Suppose that $v_2 = v_3$. Consider two estimators:

3-point average: $M_3 = \frac{1}{3}(v_1 + v_2 + v_3)$

2-point average: $M_2 = \frac{1}{2}(v_1 + v_2)$

Monte Carlo Estimation - non-IID Case

Note that

$$M_3 = \frac{1}{3}(v_1 + v_2 + v_3) = \frac{1}{3}v_1 + \frac{2}{3}v_2.$$

Variance of the two estimators

$$\mathbf{Var}(M_3) = \frac{5}{9} > \frac{1}{2} = \mathbf{Var}(M_2)$$

- We get worse results by using more samples!!
- Sample 3 provides no new information.
- We can find better weights than using the average.
- The optimal weights are $w_1 = 1/2$ and any w_2 and w_3 such that $w_2 + w_3 = 1/2$.

Monte Carlo Estimation - non-IID Case

Example 2 from Bersekas & Tsitkil p.183

Suppose that we have $2N$ samples

$$v_1, \bar{v}_1, v_2, \bar{v}_2, \dots, v_N, \bar{v}_N \sim \rho$$

with mean m and variance 1. The pairs v_k and $\bar{v}_k$ are dependent, otherwise all samples are independent. Consider two estimators:

Full-Average:
$$M_{2N} = \frac{1}{2N} \sum_{k=1}^N (v_k + \bar{v}_k)$$

Disregard dependent samples:
$$M_N = \frac{1}{N} \sum_{k=1}^N v_k$$

Monte Carlo Estimation - non-IID Case

Is it good to disregard dependent samples?

$$\mathbf{Var}(M_N) = \frac{1}{N} \quad \text{and} \quad \mathbf{Var}(M_{2N}) = \frac{1 + \rho}{2N}$$

where $\rho = \mathbf{Cov}(v_k, \bar{v}_k)$. Since $\rho \leq 1$, the estimators are in the worst case equal.

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Value Estimation

Consider a shortest-path MRP $(\mathcal{S}, P, R, \mu)$:

- 1 **States:** $\mathcal{S} = \{0, 1, 2, \dots, n\}$
- 2 **Transitions:** $P : \mathcal{S} \times \mathcal{S} \rightarrow [0, 1]$ where

$$\sum_{s' \in \mathcal{S}} P(s, s') = 1 \quad \text{for all } s \in \mathcal{S}.$$

- 3 **Reward function:** $R : \mathcal{S} \rightarrow \mathbb{R}$
- 4 **Terminal state:** 0 is a reward free terminal state, i.e.
 $R(0) = 0$ and $P(0, 0) = 1$ and $P(0, s) = 0$ for all $s \in \mathcal{S} \setminus \{0\}$.
- 5 **Value function:**

$$V(s) = \mathbf{E} \left[\sum_{t=0}^{\infty} R(S_t) \mid S_0 = s \right]$$

Monte Carlo Value Estimation

Suppose that we run $e = 1, 2, \dots, E$ independent episodes

Episode 1: $S_0^1, R_0^1, S_1^1, R_1^1, \dots, S_{T_1}^1, R_{T_1}^1$

Episode 2: $S_0^2, R_0^2, S_1^2, R_1^2, \dots, S_{T_1}^2, R_{T_1}^2$

$\vdots$

Episode E : $S_0^E, R_0^E, S_1^E, R_1^E, \dots, S_{T_E}^E, R_{T_E}^E$

where $R_t^e = r(S_t^e)$.

Stopping time

We stop each episode e when we reach the terminal state 0:

$$T_e = \min\{t \in \mathbb{N} | S_t^e = 0\}.$$

Monte Carlo Value Estimation

Problem

Suppose that we don't know P and R . How can we estimate the value function $V(s)$ from the sample episodes?

Monte Carlo Value Estimation

Simple case

Suppose that $\mathbf{P}[S_0 = s] = 1$ for some $s \in \mathcal{S}$. How can we estimate $V(s)$?

Define

$$C^e(s) = \sum_{t=0}^{T^e} R_t^e, \quad \text{for episodes } e = 1, 2, 3, \dots, E$$

Then C^e is a random variable. Is it easy to show that to establish that

$$V(s) = \mathbf{E}[C^e(s)]?$$

Monte Carlo Value Estimation

Consider the estimator

$$\hat{V}(s) = \frac{1}{E} \sum_{e=1} C^e(s)$$

Is it consistent?

What could be the problem?

Monte Carlo Value Estimation

Assumption: Proper MRP

We say that the MRP is **proper** if

$$\rho := \max_{s=1,2,\dots,n} \mathbf{P}(S_n \neq | S_0 = s) < 1.$$

- If the MRP is proper then we can prove that (Homework 3)

$$\mathbf{Var}(C^e(s)) \leq \sigma^2$$

for some σ^2 .

- If the episodes are statistically independent then $\hat{V}(s)$ is consistent.

Monte Carlo Value Estimation

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Monte Carlo Value Estimation

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Why Do We Need Importance Sampling?

Suppose we want to estimate

$$\mathbb{E}_{X \sim p}[f(X)]$$

but our samples come from another distribution

$$X_1, \dots, X_N \sim q.$$

Question: Can we still estimate the expectation under p ?

Importance Sampling

Assume $q(x) > 0$ whenever $p(x) > 0$.

$$\begin{aligned}\mathbb{E}_p[f(X)] &= \sum_x p(x)f(x) \\ &= \sum_x q(x) \frac{p(x)}{q(x)} f(x) \\ &= \mathbb{E}_q \left[\frac{p(X)}{q(X)} f(X) \right].\end{aligned}$$

Define the importance weight

$$w(X) = \frac{p(X)}{q(X)}.$$

Importance Sampling Estimator

Given samples

$$X_1, \dots, X_N \sim q,$$

consider

$$\hat{\mu}_N = \frac{1}{N} \sum_{k=1}^N w(X_k) f(X_k).$$

Then

$$\mathbb{E}[\hat{\mu}_N] = \mathbb{E}_p[f(X)].$$

Importance sampling gives an unbiased estimator.

Importance Sampling in RL

Target policy:

$$\pi$$

Behavior policy:

$$\mu$$

Data are generated using μ , but we want the value of π . For a trajectory

$$\tau = (s_0, a_0, s_1, a_1, \dots),$$

the importance ratio is

$$W(\tau) = \prod_t \frac{\pi(a_t|s_t)}{\mu(a_t|s_t)}.$$

$$V^\pi(s) = \mathbb{E}_\mu[W(\tau)G(\tau)].$$

Limitations of Monte Carlo Methods

Monte Carlo methods:

- require complete trajectories,
- may have high variance,
- importance sampling can further increase variance.

Can we update values after every transition?

⇒ Temporal Difference Learning

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TD(0)

Consider a discounted Markov reward process. After observing

$$S_k \xrightarrow{R_{k+1}} S_{k+1},$$

the TD(0) update is

$$V_{k+1}(S_k) = V_k(S_k) + \alpha_k \left(R_{k+1} + \gamma V_k(S_{k+1}) - V_k(S_k) \right).$$

All other states remain unchanged.

The Bellman Operator

Define

$$(TV)(s) = r(s) + \gamma \sum_{s'} P(s, s') V(s').$$

The value function is the unique solution of

$$TV = V.$$

Goal:

$$\text{solve } TV = V$$

using only simulated transitions.

A Sample of the Bellman Operator

Conditioned on the current state,

$$\mathbb{E}\left[R_{k+1} + \gamma V_k(S_{k+1}) \mid S_k = s\right] = (TV_k)(s).$$

Thus

$$R_{k+1} + \gamma V_k(S_{k+1})$$

is an unbiased sample of

$$(TV_k)(S_k).$$

Bellman Error + Noise

Define

$$M_{k+1} = R_{k+1} + \gamma V_k(S_{k+1}) - (TV_k)(S_k).$$

Then

$$\mathbb{E}[M_{k+1} \mid \mathcal{F}_k] = 0.$$

Therefore

$$R_{k+1} + \gamma V_k(S_{k+1}) = (TV_k)(S_k) + M_{k+1}.$$

Hence

$$\delta_k = (TV_k)(S_k) - V_k(S_k) + M_{k+1}.$$

TD as Stochastic Approximation

The update becomes

$$V_{k+1}(S_k) = V_k(S_k) + \alpha_k \left((TV_k)(S_k) - V_k(S_k) + M_{k+1} \right).$$

Ignoring the asynchronous updates,

$$V_{k+1} = V_k + \alpha_k \left(TV_k - V_k + M_{k+1} \right).$$

This is of the form

$$x_{k+1} = x_k + \alpha_k (F(x_k) + M_{k+1}),$$

with

$$F(V) = TV - V.$$

Mean Dynamics

Ignoring the noise gives

$$V_{k+1} = V_k + \alpha_k(TV_k - V_k).$$

Equivalently,

$$\dot{V} = TV - V.$$

Question: To what point does this dynamical system converge?

Fixed Points

A fixed point satisfies

$$TV - V = 0.$$

Hence

$$TV = V.$$

But this is precisely the Bellman equation. Therefore the equilibrium is

$$v.$$

Why is the Equilibrium Stable?

The Bellman operator satisfies

$$\|TV - TW\|_{\infty} \leq \gamma \|V - W\|_{\infty}, \quad 0 \leq \gamma < 1.$$

Therefore T is a contraction and

$$TV = V$$

has a unique solution v . Hence

$$\dot{V} = TV - V$$

has a globally asymptotically stable equilibrium.

A General Stochastic Approximation Result

Consider

$$x_{k+1} = x_k + \alpha_k(F(x_k) + M_{k+1}).$$

Assume:

- $\sum_k \alpha_k = \infty$,
- $\sum_k \alpha_k^2 < \infty$,
- M_{k+1} is a martingale difference noise,
- $\dot{x} = F(x)$ has a globally asymptotically stable equilibrium.

Then

$$x_k \rightarrow x^* \quad \text{a.s.}$$

Convergence of TD(0)

We have shown:

$$F(V) = TV - V.$$

The noise satisfies

$$\mathbb{E}[M_{k+1}|\mathcal{F}_k] = 0.$$

The ODE

$$\dot{V} = TV - V$$

has the globally asymptotically stable equilibrium

$$v.$$

Therefore

$$V_k \rightarrow v \quad \text{a.s.}$$

Fixed Step-Size TD

Consider TD with a constant step size:

$$V_{k+1} = V_k + \alpha(TV_k - V_k + M_{k+1}).$$

where

$$0 < \alpha \ll 1.$$

Question:

How close does V_k get to v ?

Error Dynamics

Let

$$v = Tv$$

denote the true value function and define

$$e_k = V_k - v.$$

Since

$$T(V) = r + \gamma PV,$$

we have

$$T(V) - T(v) = \gamma P(V - v).$$

Therefore

$$e_{k+1} = (I - \alpha(I - \gamma P))e_k + \alpha M_{k+1}.$$

A stable linear system driven by noise.

Convergence Rate

Ignoring the noise,

$$e_{k+1} = (I - \alpha(I - \gamma P))e_k.$$

The eigenvalues satisfy

$$|\lambda_i(P)| \leq 1.$$

Hence

$$|\lambda_i(I - \gamma P)| \leq 1 - \gamma.$$

Therefore

$$\|e_k\| \lesssim (1 - \alpha(1 - \gamma))^k \|e_0\|.$$

To get $\epsilon > 0$ accuracy:

$$k_\epsilon = O\left(\frac{\log(1/\epsilon)}{\alpha(1 - \gamma)}\right).$$

Steady-State Variance

Let

$$\Sigma = \lim_{k \rightarrow \infty} \mathbb{E}[e_k e_k^\top].$$

The covariance satisfies

$$\Sigma = A \Sigma A^\top + \alpha^2 Q,$$

where

$$A = I - \alpha(I - \gamma P).$$

For small α ,

$$\Sigma = O(\alpha).$$

Consequently,

$$\mathbb{E}\|e_k\|^2 = O(\alpha).$$

Initial Error and Noise

Since

$$\|A\| \leq 1 - \alpha(1 - \gamma),$$

the initial error decays as

$$\|A^k e_0\| \leq (1 - \alpha(1 - \gamma))^k \|e_0\|.$$

The noise contribution satisfies

$$\alpha^2 \sum_{i=0}^{k-1} \|A\|^{2i} \mathbb{E} \|M_{k-i}\|^2.$$

Using

$$\|A\| \leq 1 - \alpha(1 - \gamma),$$

gives

$$\alpha^2 \sum_{i=0}^{\infty} (1 - \alpha(1 - \gamma))^{2i}.$$

Asymptotic Error

The geometric series evaluates to

$$\sum_{i=0}^{\infty} (1 - \alpha(1 - \gamma))^{2i} = O\left(\frac{1}{\alpha(1 - \gamma)}\right).$$

Hence

$$\mathbb{E}\|e_k\|^2 = O\left(\frac{\alpha}{1 - \gamma}\right).$$

Equivalently,

$$\sqrt{\mathbb{E}\|e_k\|^2} = O\left(\sqrt{\frac{\alpha}{1 - \gamma}}\right).$$

Smaller step sizes reduce the asymptotic fluctuations.

The Fixed Step-Size Tradeoff

As $k \rightarrow \infty$,

$$\mathbb{E} \|V_k - v\|^2 = O\left(\frac{\alpha}{1-\gamma}\right)$$

while

$$k_{\text{mix}} = O\left(\frac{1}{\alpha(1-\gamma)}\right)$$

Large α :

- faster learning,
- larger fluctuations.

Small α :

- slower learning,
- more accurate estimates.